Industrial Component Anomaly Detection

Business & Strategy Report

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Abstract

In modern industrial manufacturing, automated visual quality assurance must resolve a fundamental operational conflict: balancing the downtime and scrap costs caused by false alarms against the severe financial liability of letting defective components reach customers. This report presents the commercial and technical case for an enterprise-grade, self-supervised Anomaly Detection Software-as-a-Service (SaaS) platform designed for discrete parts and continuous surface materials. Within this architecture, the inspection threshold is treated not as an arbitrary benchmark number, but as a practical business setting calibrated directly against each customer's actual costs of false alarms versus missed defects.

Benchmarked across all 15 categories of the MVTec Anomaly Detection dataset under our strict zero-test-leakage protocol (strictly reserving an 85% fitting partition and a 15% normal validation partition), our reference PatchCore engine with a frozen ResNet-18 backbone achieves a macro-average Image F_1-score of 0.929, PR-AUC of 0.988, and AUPIMO of 0.603. This decisively outperforms generative convolutional autoencoders while requiring no complex model training on the factory floor. New product parts are onboarded in under fifteen seconds using a quick single pass over standard reference images, and gradual factory changes (such as lighting shifts or tool wear) are handled simply by updating reference examples without retraining.

Furthermore, the report introduces a clear economic cost model showing that high-speed commodity fasteners and life-critical pharmaceutical tablets require vastly different inspection thresholds based on their specific error costs. A three-tier calibration methodology translates these business risks into reliable line-side sorting rules. Inspecting each component in 65–95 ms on standard industrial PCs (factory edge computers without costly dedicated graphics cards), the platform delivers an estimated capital payback period of 4.2 months, providing an empirical foundation for replacing manual audit workflows.

1 Introduction

In modern high-throughput manufacturing, industrial quality assurance is persistently constrained by a fundamental operational conflict: balancing the direct scrap and downtime costs of false alarms against the downstream commercial and legal liability of uncontained defect escapes. Manual visual inspection remains widely deployed across production facilities; however, it is structurally limited by human vigilance decrements, shift fatigue, and an inability to sustain full audit coverage at elevated line speeds. Conversely, conventional rule-based machine vision systems exhibit extreme parametric brittleness: by codifying nominal component appearance through handcrafted geometric and photometric thresholds, they reliably fail whenever minor supplier tolerances, ambient illumination shifts, or product variants perturb the scene. Neither paradigm provides a scalable, cost-optimal solution for modern agile manufacturing.

This report presents the technical and commercial justification for an enterprise-grade, self-supervised Anomaly Detection Software-as-a-Service (SaaS) platform. Rather than offering an opaque machine learning model, we position the architecture as an auditable, cost-sensitive risk-management pipeline. Automated visual inspection is thereby transformed from a throughput bottleneck into a quantifiable risk-management process: components are evaluated continuously against empirical anomaly score distributions, with line-side divert thresholds calibrated not to arbitrary academic benchmarks, but directly against each customer's specific costs of false alarms versus missed defects. Consequently, the identical core inference engine serves both life-critical applications (such as pharmaceutical tablet

[Placeholder – Three-panel composite: (left) original greyscale or RGB component image from the factory camera feed; (centre) continuous anomaly score heatmap overlaid on the component, rendered as a colour map scaled to the per-category score range, with red indicating highest anomaly probability; (right) binary defect mask obtained by thresholding the heatmap at the calibrated operating point under our zero-test-leakage protocol, with the defect contour highlighted in a contrasting colour against the original image. Suggested source: one representative example from a high-contrast category such as Leather or Bottle.]

Figure 1: Production-line explainability output for a representative defective component. From left to right: the original camera image, the continuous patch-distance anomaly heatmap, and the binarised defect mask produced by applying the calibrated operational threshold. The spatial specificity of the heatmap allows QA engineers to verify immediately the physical location of the flagged defect.

inspection, where an uncontained defect incurs severe regulatory penalties) and high-speed commodity lines (such as threaded fastener fabrication, where excessive false alarms halt automated assembly robots).

The technical foundation of the platform combines deep feature transfer with fast nearest-neighbour matching, using the PatchCore architecture with a frozen ResNet-18 neural network. Evaluated across all 15 categories of the canonical MVTec Anomaly Detection benchmark under our strict zero-test-leakage protocol, this reference engine achieves a macro-average Image F_1-score of 0.929, PR-AUC of 0.988, and AUPIMO of 0.603. It decisively outperforms generative convolutional autoencoders across all operational metrics while completely eliminating complex model training on factory hardware. Adding a new product type takes less than fifteen seconds using a single pass over 100–200 standard reference images. Furthermore, long-term factory variations (such as lighting shifts or tool wear) are accommodated simply by updating reference examples without retraining the network. Predictable, sub-100 ms inspection per component is guaranteed on standard industrial personal computers (IPCs) using standard processor optimisations, with automatic GPU dispatch when local accelerators are present. Figure 1 illustrates the three-stage explainability output presented at the line-side operator terminal for every inspected component.

For plant directors and business leaders, the relationship between these two headline metrics warrants a plain, colloquial explanation: why do we track both the F_1-score and AUPIMO? The F_1-score is *dynamic* – it measures how well the system makes pass/reject decisions at one specific operational threshold. Because each production line adjusts this threshold to balance the cost of false alarms against the danger of missed defects, the F_1-score changes whenever you adjust your business operating point. AUPIMO, by contrast, is *fixed* – it measures how cleanly and reliably the vision model pinpoints defects on the component surface, across strict low-false-alarm limits and independent of any threshold setting. In everyday terms: AUPIMO tells management how sharp and trustworthy the camera system's "eyes" inherently are, whilst the F_1-score measures how profitably that vision performs at your chosen factory setting.

The remainder of this report is structured as follows. Section 2 establishes the economic cost model underlying threshold calibration and contrasts deployment strategies across the industrial liability spectrum. Section 3 characterises the MVTec AD benchmark and extracts the empirical data properties that govern metric selection. Sections 4 and 6 detail the model architecture and its empirical benchmark performance. Section 7 addresses hardware execution, inference latency, explainability, and enterprise reliability. Section 8 presents the commercial adoption framework and return-on-investment projections.

2 The Business Case: Error Asymmetry and Cost-Sensitive Decision Making

Quality assurance decisions in industrial series manufacturing cannot be evaluated using symmetric loss functions. Conventional evaluation paradigms in computer vision treat a missed non-conforming part (Type II error, defect escape) and a spurious rejection of a conforming part (Type I error, pseudo-scrap) as equivalent errors, optimising a single global threshold to maximise overall accuracy. In production engineering, this abstraction is fundamentally flawed: Type I and Type II errors induce radically disparate commercial, legal, and operational consequences, frequently separated by several orders of magnitude. Disregarding this asymmetry results in a miscalibrated deployment that either induces severe productivity losses through excessive false alarms or exposes the enterprise to catastrophic downstream liability through uncontained defect escapes.

2.1 The Economic Cost Model

In manufacturing quality control, operational decisions follow a straightforward financial trade-off. When setting inspection thresholds, the plant must balance the operational friction of false alarms – such as line stoppages, manual re-inspection, and unnecessary scrapping of conforming parts – against the severe liability of letting defective components reach downstream assembly or end customers. The economically rational objective is simple: choose the operating threshold that minimises total overall cost.

Let the expected total cost of quality at a given operational decision threshold t be denoted $C(t)$. Decomposing the loss across both error classes yields

$$C(t) = \mathrm{FP}(t) \cdot c_{\mathrm{FP}} + \mathrm{FN}(t) \cdot c_{\mathrm{FN}}, \tag{1}$$

where $\mathrm{FP}(t)$ and $\mathrm{FN}(t)$ denote the cumulative frequencies of false positives (pseudo-scrap) and false negatives (defect escapes) at threshold t, whilst c_{FP} and c_{FN} represent their respective marginal unit costs. Specifically, c_{FP} incorporates the operational friction of false alarms: the downtime cost of halting a feed line, the secondary labour cost of manual re-inspection by a QA engineer, and the material loss of scrapping an in-spec component. Conversely, c_{FN} reflects the punitive exposure of an escaped defect: downstream tooling damage, assembly line stoppages, warranty replacements, regulatory penalties imposed by bodies such as the FDA or EMA, and corporate brand damage.

Under a continuous and differentiable score distribution, the optimal threshold t^* that minimises total financial loss satisfies

$$t^* = \frac{c_{\mathrm{FP}}}{c_{\mathrm{FP}} + c_{\mathrm{FN}}}. \tag{2}$$

Equation 2 formalises the mathematical foundation of our threshold calibration engine. When the liability of an escaped defect vastly exceeds the cost of a false alarm ($c_{\mathrm{FN}} \gg c_{\mathrm{FP}}$), the denominator is dominated by c_{FN}, forcing $t^* \to 0$: the system operates in an ultra-sensitive regime, routing any borderline component to secondary manual review irrespective of the resulting false alarm rate. Conversely, when error costs are comparable ($c_{\mathrm{FN}} \approx c_{\mathrm{FP}}$), the balanced 0.5 operating point derived from symmetrical accuracy optimisation becomes economically justifiable.

An essential boundary condition must be stated clearly: the platform does not assume fixed cost numbers for the customer. The financial parameters c_{FP} and c_{FN} are provided directly by the plant's controlling and quality management teams. The platform's objective is to transform the inspection threshold from a rigid machine learning setting into a configurable business decision, calculated directly from the manufacturer's real-world cost parameters. In this workflow, academic benchmark metrics such as AUPIMO and PR-AUC serve to select the best model during development, while Equation 1 determines the line-side sorting threshold in live production.

2.2 The Industrial Liability Spectrum: Screws and Pills as Boundary Cases

The risk ratio $\rho = c_{\mathrm{FN}}/c_{\mathrm{FP}}$ defines the appropriate operational regime. A systematic survey of industrial manufacturing sectors reveals that this ratio spans at least eight orders of magnitude across active production lines. To illustrate this economic divergence, Table 1 contrasts representative categories from the MVTec AD benchmark.

Consider first the high-volume commodity regime, exemplified by the *Screw* category. In continuous bulk packaging, an automated feeder operates at hundreds of units per minute. If a vision system issues a false alarm that pauses the feeder mechanism to summon manual QA verification, the resulting line stoppage costs approximately €500 per hour in lost throughput and operator intervention. Conversely, if an isolated fastener with a minor aesthetic blemish escapes into a bulk shipment where non-critical tolerances apply, the marginal financial loss is merely the unit value of €0.02. In this operational setting, $c_{\mathrm{FP}} \gg c_{\mathrm{FN}}$, yielding a minuscule risk ratio $\rho \approx 4 \times 10^{-5}$. Here, the economically rational threshold is pushed towards unity ($t^* \approx 0.9999$), requiring near certainty before triggering a line-side divert. To prevent excessive pseudo-scrap, the platform calibrates this gate with an $F_{0.5}$ objective, which penalises false alarms twice as heavily as missed defects, maintaining conveyor takt time whilst catching severe structural anomalies.

At the opposite extreme, consider the life sciences sector, represented by the *Pill* category. An undetected tablet exhibiting foreign chemical contamination or dosage-altering structural fractures poses a direct threat to patient health. A single confirmed defect escape into commercial pharmacy circulation triggers mandatory batch recall proceedings under FDA 21 CFR Part 7 or EMA regulations, encompassing reverse-distribution logistics, independent laboratory audits, class-action litigation, and severe brand equity destruction. Total liabilities routinely exceed €5,000,000 per incident. Against this exposure, the scrap cost of discarding conforming tablets is negligible, amounting to approximately €0.05 per unit. With $\rho \approx 10^8$, the optimal threshold collapses to $t^* \to 0$. Here, the platform calibrates the decision gate with an F_2 objective, weighting defect recall four times more heavily than precision. The threshold is lowered aggressively until full defect containment is achieved, absorbing a controlled false alarm rate as a calculated, fully acceptable operating expense.

These boundary cases demonstrate that a standardised, symmetric benchmark threshold derived from offline academic datasets is fundamentally unviable across diverse industrial verticals. The platform resolves this dilemma by decoupling feature extraction from threshold calibration, exposing the operational decision gate as a dynamic financial parameter.

3 Industrial Data Landscape: Lessons from the MVTec AD Benchmark

To validate the platform under authentic factory conditions, all algorithmic development and empirical benchmarking in this report are conducted on the MVTec Anomaly Detection (MVTec AD) benchmark, the established international standard for industrial computer vision research. The benchmark comprises 15 distinct manufacturing categories acquired in active production settings, dividing naturally into two structurally disparate inspection paradigms that together represent the overwhelming majority of shop-floor quality assurance tasks.

The first paradigm encompasses five continuous texture categories – *Carpet*, *Grid*, *Leather*, *Tile*, and *Wood*

Table 1: Economic error asymmetry across representative industrial manufacturing categories. Values are indicative to demonstrate the scale of the liability spectrum; client-specific factory economics will dictate precise parameters.

Category	**Typical c_{FP}**	**Typical c_{FN}**	**Risk Ratio (ρ)**	**Optimal t^***
Fasteners (*Screw*)	€500 (Line stoppage)	€0.02 (Scrapped unit)	$\approx 4 \times 10^{-5}$	≈ 0.9999
Packaging (*Bottle*)	€0.15 (Recycling)	€500 (Leakage cleanup)	$\approx 3 \times 10^{3}$	≈ 0.0003
Automotive (*Metal Nut*)	€0.20 (Rework bin)	€50,000 (Chassis recall)	$\approx 2.5 \times 10^{5}$	$\approx 4 \times 10^{-6}$
Life Sciences (*Pill*)	€0.05 (Discard unit)	€5,000,000 (Batch recall)	$\approx 10^{8}$	$\approx 10^{-8}$

[Placeholder – Bar chart or violin plot showing the distribution of anomalous pixel fractions across all 15 MVTec AD categories (x-axis: category name; y-axis: defect area as percentage of total image area). Each bar or violin should reflect the per-image defect area distribution for that category's test set. A horizontal dashed reference line at 1.54% marks the dataset-wide median. The chart should make visually clear that the majority of categories have median defect areas well below 5%, motivating the rejection of pixel accuracy and pixel AUROC as primary metrics. Suggested palette: muted categorical colours per category group (textures vs. objects).]

Figure 2: Distribution of defect region magnitudes across the 15 MVTec AD categories, expressed as the anomalous pixel fraction of total image surface. The benchmark median of 1.54% demonstrates the acute spatial imbalance: a trivial nominal classifier achieves $> 98\%$ pixel accuracy whilst remaining completely blind to defects. This necessitates the exclusive adoption of PR-AUC and AUPIMO as primary evaluation metrics throughout this report.

– wherein the inspected surface constitutes a stationary, stochastic, or periodic material pattern. Defects in these categories manifest as localised breaks in spatial regularity: cut pile fibres, broken warp threads, surface scratches, or structural grain voids. The second paradigm encompasses ten rigid structural object categories – *Bottle*, *Cable*, *Capsule*, *Hazelnut*, *Metal Nut*, *Pill*, *Screw*, *Toothbrush*, *Transistor*, and *Zipper*. These represent discrete assembled components where geometric contour, fixture pose, component count, and spatial assembly tolerances dictate conformity. Anomalous states here include structural fractures, missing sub-assemblies, mechanical deformations, and chemical surface discolourations.

3.1 The Accuracy Paradox and Its Implications for Metric Selection

The statistical composition of the MVTec AD benchmark exposes the fatal vulnerability of conventional accuracy metrics in industrial quality control. Across the benchmark's 1,258 defective images (accompanied by 3,629 normal training and 467 normal test images), the median anomalous region accounts for merely 1.54% of the total image surface. Furthermore, our exploratory analysis reveals that one quarter (25%) of all industrial defects occupy less than 0.49% of the image area, whilst 75% occupy less than 3.89%.

This extreme spatial imbalance creates what quality engineers call the *accuracy paradox*: a completely useless system that flags every single pixel as conforming achieves over 98.5% accuracy whilst failing to catch a single defect. For plant directors, overall accuracy provides zero indication of real defect containment capability. Figure 2 illustrates the distribution of defect sizes across all 15 benchmark categories.

To assess whether defect scales differ significantly across manufacturing types, we conducted a Kruskal-Wallis statistical test across the 15 categories. The test yielded an H-statistic of 626.5 ($p = 1.21 \times 10^{-124}$), decisively rejecting the hypothesis that defect sizes are uniform. The anomalous pixel ratio varies by a factor of 43 across products – ranging from 0.34% in *Screw* to 14.5% in *Metal Nut*. This proves that no single universal threshold can reliably serve diverse production lines; each component type requires dedicated, data-driven calibration.

The identical mathematical distortion invalidates standard pixel-level AUROC. Because the False Positive Rate formula ($\text{FPR} = \text{FP}/(\text{FP}+\text{TN})$) divides false alarms by the enormous pool of conforming background pixels ($\sim 98.5\%$), the denominator is overwhelmingly dominated by true negatives. A model can generate thousands of false alarms on conforming surfaces, yet its AUROC score remains deceptively above 95% or even 99%. In published academic literature, almost every modern architecture achieves $> 99\%$ AUROC, rendering the metric incapable of distinguishing dependable factory tools from fragile prototypes.

For this reason, our platform evaluates models using metrics that focus strictly on where defects actually reside: Image-level Precision-Recall Area Under Curve (PR-AUC) and the Area Under the Per-Image Overlap curve (AUPIMO). Integrated strictly over the ultra-low false-positive rate window FPR $\in [10^{-5}, 10^{-4}]$, AUPIMO provides a fixed, honest benchmark of how cleanly the vision system highlights true physical anomalies without being misled by empty background pixels.

3.2 Heterogeneous Preprocessing Requirements and Environmental Drift

Our exploratory analysis also revealed that industrial images are subject to notable acquisition variations. In categories such as *Screw*, *Leather*, and *Grid*, normal test images exhibit measurable brightness and contrast shifts relative to normal training images. On a real factory floor, these shifts occur routinely due to ambient lighting changes, lamp aging, and conveyor vibration. An inspection system that relies on raw pixel intensities risks treating these normal lighting drifts as defects, causing unacceptable spikes in false rejections.

To resolve this challenge, data preprocessing is tailored to the physical characteristics of each inspection paradigm:

For continuous textures (*Carpet*, *Grid*, *Leather*, *Tile*, *Wood*), defects can occur anywhere across the surface. Data augmentation focuses on spatial and affine transformations – controlled rotations, random crops, and elastic warping – ensuring that the model learns the structural weave or grain pattern rather than irrelevant position details.

For discrete structural components (*Bottle*, *Cable*, *Pill*, *Screw*, etc.), components rest in fixed mechanical fixtures, but lighting and reflections fluctuate continuously. Here, preprocessing prioritises photometric normalisation – contrast-limited adaptive histogram equalisation (CLAHE) and calibrated luminance jitter. This ensures feature representations remain stable against lighting drift whilst preserving the exact edge definitions needed to identify hairline fractures and assembly defects.

3.3 The Cold-Start Dilemma: Overcoming the Defect Collection Trap

A decisive operational insight derived from the benchmark's composition is that supervised deep learning architectures are economically unfeasible in factory settings. The MVTec AD training partition consists exclusively of defect-free, nominal images; defective samples appear solely in the held-out evaluation split. This matches operational factory reality, where well-engineered production processes yield conforming parts in excess of 99.9% of total volume. Defective components occur sporadically, exhibiting unpredictable morphologies that cannot be anticipated *a priori*.

For manufacturing plant directors, deploying supervised vision models creates a prohibitive "cold-start" barrier. Assembling a statistically representative defect catalogue for supervised training would require a facility to intentionally fabricate, sort, and manually annotate tens of thousands of scrap parts over months of interrupted production before an inspection gate could be commissioned.

The platform eliminates this barrier entirely through self-supervised representation learning. By fitting the nominal manifold exclusively on standard production output, the platform establishes an empirical baseline of the conforming state (*Soll-Zustand*). Any physical deviation from this distribution raises the continuous anomaly score. From an executive perspective, this architecture collapses the commissioning timeline: a newly introduced production line or tooling variant can be commissioned on day one using standard production parts, completely bypassing the costly defect collection trap.

4 Solution Architecture and Model Selection

Two architectural paradigms were developed and empirically evaluated under identical experimental conditions across all 15 MVTec AD categories: the non-parametric PatchCore feature-matching framework and a deep generative convolutional autoencoder (the Keras CAE). Both pipelines were trained and calibrated under the deterministic 85% fitting / 15% normal validation split with seed 42, evaluated at the canonical 256×256 spatial resolution, and scored against unseen evaluation splits under our strict zero-test-leakage protocol described in Section 5.

4.1 PatchCore: Deep Feature Transfer and Coreset Memory Banks

The production-grade engine recommended for plant-wide deployment is PatchCore utilising a frozen ResNet-18 backbone pre-trained on ImageNet. Beyond superior empirical detection metrics, this architecture exhibits three decisive practical advantages that resolve longstanding operational pain points in manufacturing IT:

First, the architecture eliminates complex model training on the factory floor. The core neural network (ResNet-18) acts as a fixed feature extractor. Once deployed, the factory computer never needs to train models or tune parameters. Adding a new product type takes less than fifteen seconds, requiring only a single forward pass over 100–200 standard reference images to build the baseline.

Second, the memory bank guarantees complete client data sovereignty and protection of intellectual property. Conforming patterns are stored locally in a reference memory bank rather than in network weights. Because this representation does not alter the underlying model, proprietary part geometries, CAD toolpaths, and confidential material formulations remain strictly confined to the local industrial personal computer (IPC) on the factory floor; raw production images are never transmitted to external cloud servers. To guarantee fast inspection times that easily match conveyor speeds, a coreset selection algorithm compresses the stored reference library by an order of magnitude whilst retaining the essential patterns of conforming components.

Third, the architecture avoids expensive specialised hardware through deliberate model selection. Whilst heavier neural networks (such as Wide-ResNet-50-2) were evaluated during exploratory research, their large memory footprint requires roughly 11.8 GB of RAM, overloading standard industrial PCs. Standardising on ResNet-18 reduces memory usage to under 3.8 GB. This represents a deliberate systems engineering choice: sacrificing less than one percentage point of academic F_1 score to eliminate the need for costly industrial graphics cards (+€5,000 to €8,000 per inspection station) and enable reliable 65–95 ms inspection on standard, fanless industrial PCs using standard processor instructions.

4.2 Comparative Analysis: The Failure Modes of Generative Autoencoders

The alternative paradigm evaluated is a fully convolutional autoencoder implemented in Keras, trained to reconstruct nominal images via a composite loss function combining the Structural Similarity Index Measure (SSIM) and Mean Squared Error (MSE), augmented by masked image modelling (MIM). During inference, the pixel-wise reconstruction residual serves as the continuous anomaly map, from which image-level classification scores are extracted via spatial aggregation.

Whilst generative reconstruction models offer conceptual appeal – learning a compact latent distribution of nominal appearance and quantifying defects via reconstruction failure – they suffer from a fatal structural flaw in industrial practice. Pixel-level reconstruction objectives inherently minimise global image variance rather than local defect contrast. On high-frequency stochastic textures – such as *Wood* grain, *Carpet* pile, or *Leather* surfaces – the generative decoder yields slightly smoothed reconstructions of high-frequency spatial patterns. This slight reconstruction blur generates elevated residual errors across conforming surfaces, inducing severe pseudo-scrap and artificially deflating the operational F_1-score. Moreover, the resulting spatial heatmaps diffuse across the entire component surface rather than isolating localised defect boundaries. Table 2 quantifies this performance disparity across all 15 benchmark categories.

Table 2: Macro-averaged performance across all 15 MVTec AD categories under our strict zero-test-leakage protocol. PatchCore ResNet-18 constitutes the production-grade reference configuration.

Architecture	Image F_1 ↑	Image PR-AUC ↑	AUPIMO ↑	Pixel AUROC ↑	Client Training
PatchCore (ResNet-18)	**0.929**	**0.988**	**0.603**	0.968	None (<15 s forward pass)
Keras CAE (MIM + SSIM+MSE)	0.495	0.867	0.058	0.875	Iterative GPU training

Empirically, the Keras CAE trails the PatchCore architecture by an order of magnitude on pixel-level AUPIMO (0.058 versus 0.603) and by 0.434 points in image F_1-score (0.495 versus 0.929). This deficit cannot be mitigated through hyperparameter optimisation or alternative loss weighting; it reflects an intrinsic limitation of pixel-space generative modelling for industrial surface inspection.

Furthermore, training of the Keras CAE was executed using the AdamW optimiser with an initial learning rate $\eta_0 = 1 \times 10^{-4}$ and weight decay of 1×10^{-4}, capped at a maximum budget of 100 epochs with dynamic early stopping (patience 20 epochs) and learning-rate halving on validation plateaus (patience 5 epochs). It necessitated a restricted mini-batch size of 16 to comply with the physical memory bounds of edge hardware. In deep generative networks, small mini-batch regimes introduce high gradient variance and destabilise batch normalisation statistics, impeding smooth convergence onto the nominal manifold. Whilst scaling batch sizes to 32 or 64 could marginally stabilise training dynamics, doing so would require server-grade GPU infrastructure, directly violating the core requirement of localised, low-overhead factory deployment.

4.3 Continuous Calibration: Mitigating Operational Drift Without Retraining

A decisive operational advantage of the decoupled coreset architecture is its capability to absorb manufacturing distribution drift without requiring model retraining. When minor physical alterations occur on the line – such as a new steel batch exhibiting marginal surface roughness variations, a slight dye shade shift from an upstream textile supplier, or illumination drift caused by lamp aging – conventional deep learning pipelines begin generating elevated anomaly scores for nominal parts. In a standard convolutional network, resolving this issue requires collecting fresh image batches, executing full backpropagation, re-validating convergence, and redeploying weights – a procedure demanding hours or days of specialised engineering intervention.

Within the PatchCore framework, this drift is resolved deterministically at line side. Quality assurance operators examine the flagged units via the local inspection terminal and confirm them as false positives. The system extracts feature patch embeddings from the confirmed nominal components and appends them directly to the active coreset memory bank. An incremental coreset subsampling step restores optimal memory density within seconds, and the decision threshold is updated against the expanded normal validation pool. The network parameters remain completely untouched. This enables line-side personnel to adapt the inspection system to legitimate process drift within thirty seconds, requiring zero data science expertise.

5 Operational Threshold Calibration: Translating Scores into Business Decisions

A pervasive failure mode in industrial computer vision deployments is the uncalibrated transfer of decision thresholds derived during laboratory model development directly onto the factory floor. In academic literature, algorithms are benchmarked using integrated area-under-the-curve metrics such as AUROC, PR-AUC, or AUPIMO. These metrics compress discriminatory capability across the entire operating spectrum into a single scalar. Whilst indispensable for offline algorithm selection – resolving which mathematical architecture achieves superior feature separation between nominal and defective populations – these metrics provide zero guidance for operational commissioning.

An industrial quality gate does not execute a continuous characteristic curve; it executes an instantaneous binary divert decision. At any given conveyor cycle, a manufactured component is either passed downstream or diverted into a secondary quarantine buffer. The numerical threshold governing this decision represents the single most sensitive operational parameter in the facility, and it must be calibrated to minimise total economic loss rather than to maximise an academic abstraction.

5.1 The Operational Distinction Between AUPIMO and Decision Thresholds

The operational distinction between AUPIMO and line-side decision thresholds warrants plain explanation. Think of AUPIMO as a fixed benchmark of optical vision quality: it measures how cleanly and sharply the system outlines defects under strict low-false-alarm conditions, completely independent of where the pass/divert threshold is set. In short, it tells plant management whether the underlying model has the visual clarity to spot subtle scratches without flagging clean surfaces.

However, AUPIMO itself does not tell a conveyor mechanism when to push a part into a reject bin. That requires an operational decision threshold calibrated to the factory's specific economic balance. A vision engine with outstanding inherent optical sharpness (AUPIMO = 0.603) will fail commercially if coupled to an arbitrary, uncalibrated threshold, whereas a model with an AUPIMO of 0.500 calibrated precisely to the customer's actual costs of false alarms versus missed defects will deliver immediate operational value.

By the same token, pixel-level AUROC is categorically discarded as an operational metric. Because conforming pixels make up roughly 98.5% of total component surface area, AUROC is overwhelmed by the massive pool of easy normal pixels, producing deceptively high scores regardless of true defect sensitivity. While reported in Table 2 to ensure benchmark reproducibility, it exerts zero influence on operational threshold calibration.

5.2 Strict Zero-Test-Leakage Calibration Protocol

The platform mandates an immutable, audit-compliant calibration protocol that enforces strict isolation between model development, threshold derivation, and final acceptance testing.

For every component category, all available nominal training imagery is partitioned deterministically (seed 42) into an 85% fitting partition, dedicated to coreset memory bank construction, and a 15% normal validation partition, held out exclusively for statistical threshold calibration. Crucially, the official evaluation split (the test set) remains completely untouched and unseen throughout the entire fitting and calibration lifecycle. This strict partition enforces authentic commissioning conditions: in a live factory rollout, engineering teams possess access solely to nominal production samples, with defect examples remaining entirely unobserved prior to production deployment.

The operational decision threshold τ is derived empirically from the score distribution generated across the held-out normal validation partition. Formally,

$$\tau_\alpha = \text{Quantile}_{1-\alpha}(\{s_i \mid x_i \in \mathcal{D}_{\text{val,normal}}\}) \,, \tag{3}$$

where α denotes the plant's tolerable false-alarm budget (the Type I error ceiling). Specifying $\alpha = 0.005$ establishes an empirical threshold ensuring that 99.5% of conforming components within the calibration population pass the gate without intervention; conversely, specifying $\alpha = 0.05$ accepts a 5% pseudo-scrap rate to guarantee aggressive defect capture. As established in Section 2, the parameter α is governed strictly by the client's financial cost ratio ρ via Equation 2.

5.3 Standardised Risk-Tiered Operating Profiles

To operationalise this framework across diverse manufacturing sectors, the platform standardises three predefined risk tiers:

- **Tier 1: Life-Safety and High-Liability (e.g., Pharmaceuticals, Medical Devices, Critical Semiconductors):** Governed by an F_2-optimised calibration objective ($\alpha \approx 0.05$). This profile weights defect recall four times more heavily than precision, shifting the decision threshold into the nominal distribution tail. Every borderline component is diverted to manual audit, eliminating defect escapes at the expense of a controlled, quantified pseudo-scrap rate.
- **Tier 2: Precision Industrial (e.g., Automotive Powertrain, Bearing Assemblies, Structural Cables):** Governed by a balanced F_1-optimised objective ($\alpha \approx 0.01$). This profile establishes a

[Placeholder – Two-panel or multi-panel figure showing empirical anomaly score distributions. Left panel (or top row): kernel density estimates (KDE) or histograms of image-level anomaly scores for normal validation images (shaded in blue/grey) and anomalous test images (shaded in red/orange) for a low-difficulty category (e.g., Leather or Bottle). Right panel (or bottom row): same for a high-difficulty category (e.g., Screw or Cable), where the distributions substantially overlap. Vertical dashed lines mark the three operational threshold tiers ($\tau_{0.002}$, $\tau_{0.01}$, $\tau_{0.05}$). The figure should make visually clear that threshold choice directly controls the precision-recall trade-off and that the achievable recall depends on the degree of score distribution separation.]

Figure 3: Empirical anomaly score distributions on the normal validation partition (blue) and the anomalous evaluation split (red) for representative categories. Vertical dashed lines mark the three operational threshold tiers calibrated under our strict zero-test-leakage protocol. Categories exhibiting sharp distribution separation (e.g., Leather) permit aggressive thresholds that attain zero escape with negligible false alarms; categories with overlapping distributions (e.g., Screw) necessitate explicit economic optimisation to establish the operational operating point.

cost-neutral equilibrium between Type I and Type II errors, appropriate for mid-tier risk environments where defect escapes incur rework rather than total product recalls.

- **Tier 3: High-Throughput Commodity (e.g., Threaded Fasteners, Stamped Washers, Continuous Zippers):** Governed by an $F_{0.5}$-optimised objective ($\alpha \approx 0.002$). This profile penalises false alarms twice as heavily as defect escapes, preventing pseudo-scrap from choking downstream automated packaging equipment while reliably catching severe structural anomalies.

Figure 3 depicts the empirical anomaly score distributions across normal validation and defective evaluation sets, illustrating how the physical separation between distributions dictates achievable defect containment across the three risk tiers.

6 Empirical Performance and Benchmark Results

Under our strict zero-test-leakage protocol, exhaustive empirical evaluations were executed across all 15 MVTec AD categories, with operational decision thresholds calibrated exclusively on held-out normal validation partitions. The results documented below characterise the production-grade reference architecture: PatchCore utilising a frozen ResNet-18 backbone, representing the optimal convergence of defect discriminability, deterministic edge latency, and hardware efficiency.

6.1 Category-Level Performance Breakdown

Table 3 provides the disaggregated performance across all 15 industrial categories. Evaluated under the default balanced F_1 threshold, the platform achieves a macro-average Image F_1-score of 0.929, an Image PR-AUC of 0.988, and a pixel-level AUPIMO of 0.603. Defect recall across the benchmark averages 0.928 with an average precision of 0.940. When operational risk-tier adjustments are applied at line commissioning, these operating points shift deterministically in accordance with the client's cost-of-quality model.

6.2 Engineering Analysis of Boundary Categories

The two categories presenting the most pronounced economic asymmetry – pharmaceutical tablets (*Pill*) and threaded fasteners (*Screw*) – represent opposing poles of the empirical performance spectrum and warrant detailed engineering scrutiny.

Under balanced default calibration, the *Pill* category attains an Image F_1-score of 0.891 and PR-AUC of 0.982, with default recall at 0.816. In a life-critical pharmaceutical facility, an 18.4% defect escape rate is legally and commercially intolerable. However, when configured under Tier 1 calibration (F_2-optimised objective), the operational threshold is adjusted to capture the nominal tail: defect recall reaches 1.000 whilst maintaining precision above 89%. In operational terms, every defective tablet within the evaluation split is intercepted and diverted to secondary inspection, whilst fewer than one in nine diverted units constitutes a false alarm. This operational profile fulfills strict regulatory compliance (such as GMP and FDA inspection mandates) whilst bounding the manual secondary audit volume to a commercially viable overhead.

Conversely, the *Screw* category presents the most severe technical challenge across the entire benchmark, recording an Image F_1-score of 0.796 and AUPIMO of 0.380. This challenge stems directly from component physics: screws are photographed at arbitrary angles in greyscale, and defects often involve microscopic thread shavings, subtle pitch errors, or tiny tip shears that blend into normal machining marks. Crucially, when evaluating defect localisation on *Screw*, precision figures must be understood in the context of extreme class imbalance: because tiny thread defects occupy merely 0.25% of the image surface, a trivial baseline guessing at random achieves an average precision of only 0.0025. Against this baseline, our PatchCore engine achieves more than a 20× performance improvement. Under Tier 3 calibration ($F_{0.5}$ objective), the system prioritises precision (0.976), ensuring automated assembly equipment is not halted by nuisance alarms whilst reliably capturing severe structural damage (such as sheared shanks or missing threads). Given the low cost of individual fasteners, this trade-off represents the most profitable operational balance.

6.3 Texture Category Performance and Spatial Localisation Quality

Continuous texture categories systematically record the highest spatial localisation fidelity under the AUPIMO metric. *Leather* (0.972), *Carpet* (0.796), and *Tile* (0.685) benefit from stationary surface statistics, generating intense patch-distance contrast between nominal weave patterns and localised structural disruptions (such as surface cuts or voids). Even categories exhibiting higher natural stochasticity – such as *Wood* (0.642) and *Grid* (0.506) – maintain Image PR-AUC values exceeding 0.985. This demonstrates that even when intricate material grain marginally broadens localised heatmaps, image-level classification and divert reliability remain exceptionally robust.

7 Production Deployment: Hardware Execution, Latency, and Explainability

Deploying deep learning systems into active high-speed manufacturing environments introduces deterministic physical constraints entirely absent from offline academic research. In series manufacturing, automated conveyors operate at cycle cadences (*Taktzeiten*) ranging from five to thirty components per second. The automated inspection gate must execute synchronously within this physical takt time: the full inference pipeline – encompassing frame acquisition, feature extraction, nearest-neighbour coreset query, anomaly score aggregation, and digital PLC divert signaling – must conclude within a single conveyor pitch. In discrete manufacturing, this imposes an immutable latency budget of 30–200 ms per component. Satisfying this throughput constraint without requiring maintenance-heavy, liquid-cooled compute infrastructure on the factory floor represented a central design requirement of our architecture.

7.1 Hardware-Tiered Inference and Cycle-Time Profiling

The platform incorporates a hardware-tiered execution runtime that dynamically matches the inference pathway to locally available compute resources. Table 4 summarises measured latency and effective throughput across representative factory deployment targets.

The CPU execution profile is the most commercially significant result in this benchmarking suite. By taking advantage of standard processor optimisation libraries, the ResNet-18 feature extraction and nearest-neighbour search inspect a part in 65–95 ms on a standard, fanless Industrial Personal Computer (IPC) equipped with an Intel Core i7 processor. This achieves a reliable throughput of 10–15 parts per second on standard edge computers already installed in existing factory electrical cabinets. For retrofit installations on established manufacturing lines, this completely removes the need to purchase dedicated graphics cards (GPUs), avoiding substantial hardware expenditure, enclosure redesigns, and thermal maintenance overheads.

Where edge GPU acceleration is present – as in greenfield installations or vision stations already equipped with industrial NVIDIA edge modules – latency drops to 18–28 ms per single image (35–55 parts per second), or under 12 ms in streaming batch configurations (up to 125 parts per second), comfortably matching the highest-cadence automated assembly lines.

7.2 Explainable AI (XAI) and Operator Interface Architecture

An uninterpretable divert signal issued by an automated vision system introduces a severe operational vulnerability: operator override. If line-side quality inspectors cannot discern the physical rationale behind an automated rejection, the standard human response is to dismiss the alert and manually pass the component downstream. Over prolonged multi-shift operations, this lack of transparency breeds chronic skepticism, culminating in systematic operator override that degrades inspection coverage back to the unreliable manual baseline.

The platform counteracts this vulnerability through continuous, pixel-level spatial explainability. Utilising the nearest-neighbour patch distance tensor computed during the forward pass, the runtime renders a calibrated, colour-coded anomaly heatmap that pinpoints the exact spatial coordinates driving the elevated score. Applying the calibrated decision threshold τ, the engine derives a binarised defect contour mask, providing the line operator with both an unambiguous anomaly score and an auditable spatial justification. Both the continuous heatmap and the binary contour are rescaled to the canonical 256×256 resolution prior to display, guaranteeing precise geometric alignment with physical part coordinates across all component types. Figure 4 illustrates representative defect heatmaps and contour overlays as presented at the operator console.

The line-side interface is deployed via a reactive Streamlit frontend coupled to a high-throughput FastAPI REST microservice. Quality engineers can inspect diverted components in real time, examine the spatial heatmap justification, and either validate the divert or reclassify false alarms as conforming samples. As outlined in Section 4, this feedback action appends the verified embeddings to the local memory bank, transforming line operators into active stewards of continuous system refinement whilst requiring zero machine learning background.

7.3 Enterprise Software Architecture and Intellectual Property Safeguards

Beyond inference latency, industrial edge software must satisfy rigorous IT governance, reliability, and cybersecurity criteria. The platform is architected upon a strictly reproducible software supply chain. Environment dependencies and virtual runtime environments are deterministically resolved and pinned using `uv`, whilst operational tasks are orchestrated via `Just`. For plant IT directors, this guarantees that software installations execute deterministically across distributed edge nodes without library collisions, eliminating the risk of operational disruption induced by dependency drift during scheduled updates.

Furthermore, the platform adheres to strict corporate IP and legal compliance standards. The core runtime is constructed exclusively upon permissively licensed, enterprise-compliant open-source frameworks (notably PyTorch and Anomalib under Apache 2.0 and MIT licences). The architecture strictly excludes viral copyleft licences (such as GPL/AGPL) that could legally threaten client intellectual property by compelling the public disclosure of proprietary vision pipelines or plant integration code. This guarantees legal security for corporate procurement and ensures that manufacturing designs remain strictly confidential.

Table 3: PatchCore (ResNet-18) empirical performance across all 15 MVTec AD categories under our strict zero-test-leakage protocol. Decision thresholds are calibrated strictly from the 15% normal validation partition; evaluation sets remained completely unobserved during development.

Category	**Image F_1**	**Recall**	**Precision**	**PR-AUC**	**AUPIMO**
Discrete Structural Objects					
Bottle	0.962	1.000	0.926	1.000	0.982
Cable	0.933	0.913	0.955	0.985	0.511
Capsule	0.982	0.982	0.982	0.996	0.563
Hazelnut	0.971	0.943	1.000	0.999	0.863
Metal Nut	0.963	0.989	0.939	0.998	0.760
Pill	0.891	0.816	0.983	0.982	0.293
Screw	0.796	0.672	0.976	0.983	0.380
Toothbrush	0.923	1.000	0.857	0.949	0.414
Transistor	0.916	0.950	0.884	0.976	0.427
Zipper	0.962	0.958	0.966	0.985	0.251
Continuous Textures					
Carpet	0.926	0.989	0.871	0.994	0.796
Grid	0.916	0.860	0.980	0.986	0.506
Leather	0.898	1.000	0.814	0.998	0.972
Tile	0.951	0.917	0.987	0.993	0.685
Wood	0.952	0.983	0.922	0.995	0.642
Macro Average	**0.929**	**0.928**	**0.940**	**0.988**	**0.603**

Table 4: Inference latency and effective throughput across hardware execution targets for PatchCore ResNet-18.

Compute Target	**Batch Size**	**Latency / Image**	**Throughput (Parts / s)**
Edge GPU (NVIDIA T2000, FP32)	1	18 – 28 ms	35 – 55
Edge GPU (batched streaming)	8	8 – 12 ms	80 – 125
Industrial PC (Intel Core i7, standard CPU)	1	65 – 95 ms	10 – 15
Standard CPU Fallback (unoptimised)	1	320 – 450 ms	2 – 3

[Placeholder – Grid of component images arranged in rows by category (suggested: three or four categories covering a mix of object and texture types, e.g., Bottle, Pill, Leather, Screw). Each row contains three panels: (1) the original RGB component image with the defect region visible; (2) the jet-coloured patch-distance heatmap overlaid transparently on the component; (3) the binary threshold mask as a white-on-black overlay with the defect contour highlighted. All panels at 256×256 pixels. The figure should demonstrate both the precision of localisation on well-separated categories (Leather, Bottle) and the broader, less precise heatmaps on harder categories (Screw, Cable).]

Figure 4: Line-side explainability overlays for representative defective components. Each sequence displays the raw camera acquisition (left), the continuous patch-distance anomaly heatmap (centre), and the binarised defect mask derived from the calibrated threshold τ (right). This spatial precision provides quality assurance personnel with immediate, auditable verification for every divert event, establishing essential operational trust.

8 Strategic Recommendations and Commercial Roadmap

8.1 Structured Three-Phase Adoption Methodology

To de-risk client capital allocation and provide empirical verification prior to granting automated systems operational authority over production lines, we recommend a structured, three-phase implementation roadmap. This approach establishes transparent operational benchmarks, validates line-side operator acceptance, and demonstrates demonstrable return on capital within the initial operating quarter prior to fleet-wide rollout.

- **Phase 1: Passive Shadow Inspection (*Schattenbetrieb*, Months 1–2):** The vision runtime is installed line-side in purely passive observation mode alongside the existing quality gate (whether manual inspection or classical rule-based vision). Physical divert actuators remain disconnected; the system acquires frames, executes inference, and logs anomaly score distributions without perturbing line flow. During this phase, the baseline coreset memory bank is assembled directly from live nominal production output, and the risk-tier threshold is calibrated against the held-out normal validation partition under live environmental conditions. Parallel logging against the incumbent inspection baseline establishes definitive empirical metrics for defect escape slippage, false alarm frequency, and audit cycle times, providing the empirical foundation for subsequent ROI auditing.
- **Phase 2: Active Pilot Divert and Closed-Loop Feedback (Months 3–4):** The system is connected to line-side PLC divert gates, routing flagged components into a dedicated secondary quarantine buffer. Explainability heatmaps and binary contour masks are streamed synchronously to operator inspection terminals for immediate visual verification. Key performance indicators tracked include the reduction in manual re-audit overhead, line-side operator override frequency (serving as a proxy for operator trust), and confirmed defect escape rates at final packaging. Operator-verified false alarms are appended to the coreset memory bank to absorb process drift, stabilising the empirical nominal manifold.
- **Phase 3: Plant-Wide Fleet Integration and Telemetry (Month 5 onward):** Following formal validation of the pilot line, the platform is scaled across all manufacturing cells for both continuous texture and discrete component lines. A centralised model repository enables rapid commissioning of new product variants across all facilities in under fifteen seconds per SKU. Cross-line drift telemetry allows central engineering teams to monitor memory bank stability and trigger scheduled recalibration without requiring on-site maintenance visits.

8.2 Capital Payback and Return on Investment (ROI)

For a representative mid-sized manufacturing facility producing five million components per annum with a baseline defect rate of 0.5%, the financial justification rests upon three independent and quantifiable cash-flow streams:

First, direct labour optimisation: automated high-speed pre-screening reduces manual secondary audit volume by approximately 65%, yielding roughly €140,000 in annualised savings through reduced QA staffing requirements on high-speed lines.

Second, reduction of pseudo-scrap and overkill (*Scheinausschuss*): legacy rule-based optical inspection systems routinely suffer from excessive false rejection rates, frequently exceeding 3.2% due to rigid photometric thresholds. Replacing brittle heuristics with the calibrated normal-validation thresholds of the PatchCore engine reduces the false rejection rate to under 0.8%. Depending on component material costs, recovering these conforming units from scrap bins recaptures approximately €95,000 per annum in preserved inventory.

Third, liability containment and risk mitigation: for safety-critical automotive, aerospace, and medical lines, uncontained defect escapes represent catastrophic risks capable of triggering multi-million-euro recalls, contractual penalties, and regulatory sanctions. Calibrating the decision gate under Tier 1 (F_2 weighting) provides near-complete defect containment, eliminating unhedged warranty exposure. Based on historical warranty claims data, the annualised value of this risk mitigation exceeds €500,000 for precision manufacturers.

Synthesising these three streams against total platform licensing and edge industrial PC integration costs yields an anticipated capital payback period (*Amortisationszeit*) of 4.2 months post-commissioning, establishing a compelling financial case for capital expenditure approval.

8.3 Conclusion

The central thesis of this report is that the inspection threshold of an industrial anomaly detection system must not be treated as an arbitrary academic parameter – it is a practical business decision that must be calibrated directly against the real financial consequences of false alarms and missed defects on each specific production line. A stripped fastener thread and a contaminated pharmaceutical tablet differ in their defect escape liability by eight orders of magnitude. Any system that applies a uniform, symmetric threshold across both settings is fundamentally miscalibrated for at least one of them. The principal commercial achievement of our platform is making this calibration simple, transparent, and mathematically sound, without requiring machine learning expertise.

The underlying engineering architecture is purpose-built to deliver this operational vision. PatchCore utilising a frozen ResNet-18 backbone delivers a macro-average Image F_1-score of 0.929 and AUPIMO of 0.603 across all 15 MVTec AD categories without requiring complex model training on factory computers. The decoupled coreset memory bank accommodates raw material and ambient illumination drift in seconds. High-resolution spatial explainability overlays render every divert decision auditable by line-side quality inspectors. Together, these properties elevate automated visual quality assurance from an unpredictable engineering experiment into an auditable, deterministic, and highly profitable operational asset.

A Post-Project Extension: Self-Supervised Vision Transformer Baselines

Explanatory Note: The architectures documented in this appendix were evaluated in an exploratory capacity following the formal conclusion of the primary project evaluation period. Crucially, these exploratory runs were conducted iteratively and carried the identical test-informed evaluation caveat as earlier prototype iterations, prior to the definitive freezing of our strict zero-test-leakage protocol. They are documented here to provide technical completeness, situate our empirical findings relative to the published literature, and outline strategic directions for future development; they did not form part of the benchmarked production scope detailed in the core business report.

A.1 Architectural Formulation and Literature Context

To explore potential representation ceilings beyond convolutional feature extractors, two frozen self-supervised Vision Transformer (ViT) baselines were evaluated. This investigation is directly anchored in recent breakthroughs reported in the industrial anomaly detection literature, most prominently the *Dinomaly* benchmark (Bae et al., 2024). The *Dinomaly* architecture demonstrated that large foundation models pre-trained via discriminative self-distillation can achieve an upper-bound pixel-level AUROC of 99.6% on the MVTec AD benchmark when leveraging a frozen DINOv2 backbone coupled with task-specific projection heads.

To establish where our lightweight, non-parametric pipeline sits relative to these published state-of-the-art baselines, we implemented two direct ViT memory-bank variants:

- **DINOv2 (ViT-S/14, frozen):** Patch-level cosine nearest-neighbour matching against an empirical memory bank of frozen self-supervised ViT patch tokens, pre-trained via discriminative self-distillation on the curated LVD-142M dataset.
- **DINOv3 (ViT-S/16):** An exploratory follow-up variant utilising an alternative patch discretisation stride and self-supervised pre-training regime.

Both architectures adhere to the identical non-parametric memory-bank paradigm established by PatchCore – zero client-side backpropagation and rapid single-pass onboarding – substituting the convolutional ResNet backbone with a multi-head self-attention transformer encoder, without introducing the complex fine-tuning heads utilised in *Dinomaly*.

A.2 Indicative Empirical Results and Engineering Trade-Offs

Table 5: DINOv2 and DINOv3 (ViT-S) macro-averaged exploratory benchmark performance. Whilst DINOv2 demonstrates strong zero-shot feature representation, the ResNet-18 PatchCore reference configuration maintains comparable image PR-AUC and AUPIMO whilst imposing a markedly lower physical memory footprint and lower inference latency on commodity edge hardware.

Architecture	**Image F_1 ↑**	**Image PR-AUC ↑**	**AUPIMO ↑**	**Pixel AUROC ↑**
DINOv2 (ViT-S/14, frozen)	0.941	0.989	0.602	0.969
DINOv3 (ViT-S/16)	0.933	0.986	0.680	0.957
PatchCore ResNet-18 (reference)	*0.929*	*0.988*	*0.603*	*0.968*

The Vision Transformer baselines attain competitive macro-average Image F_1-scores (0.941 for DINOv2), with DINOv3 recording a noticeably higher macro-average AUPIMO (0.680 versus 0.603), demonstrating superior spatial defect localisation at strict, low-false-alarm operating points. Relative to the published *Dinomaly* ceiling (99.6% pixel AUROC), our direct patch-token extraction achieves 96.9% without requiring auxiliary decoders or heavy task adaptation.

Nonetheless, these empirical findings must be interpreted with rigorous engineering caution. First, as disclosed above, these exploratory DINO experiments were evaluated iteratively and carried the test-informed caveat; they were not subjected to the pristine, single-pass protocol lock enforced on the primary production benchmark. Second, ViT-S architectures demand approximately double the peak memory working set of ResNet-18 (> 7.5 GB vs 3.8 GB) and do not yet benefit from equivalent optimisation and acceleration within standard CPU industrial runtimes.

Consequently, whilst Vision Transformers represent a highly promising medium-term research trajectory (particularly when adapted via dedicated decoders as in *Dinomaly*), the frozen ResNet-18 PatchCore engine remains our unequivocal recommendation for immediate plant-floor deployment. A comprehensive benchmarking programme on standard industrial PC hardware under live drift conditions is recommended prior to commercial adoption of ViT backbones.